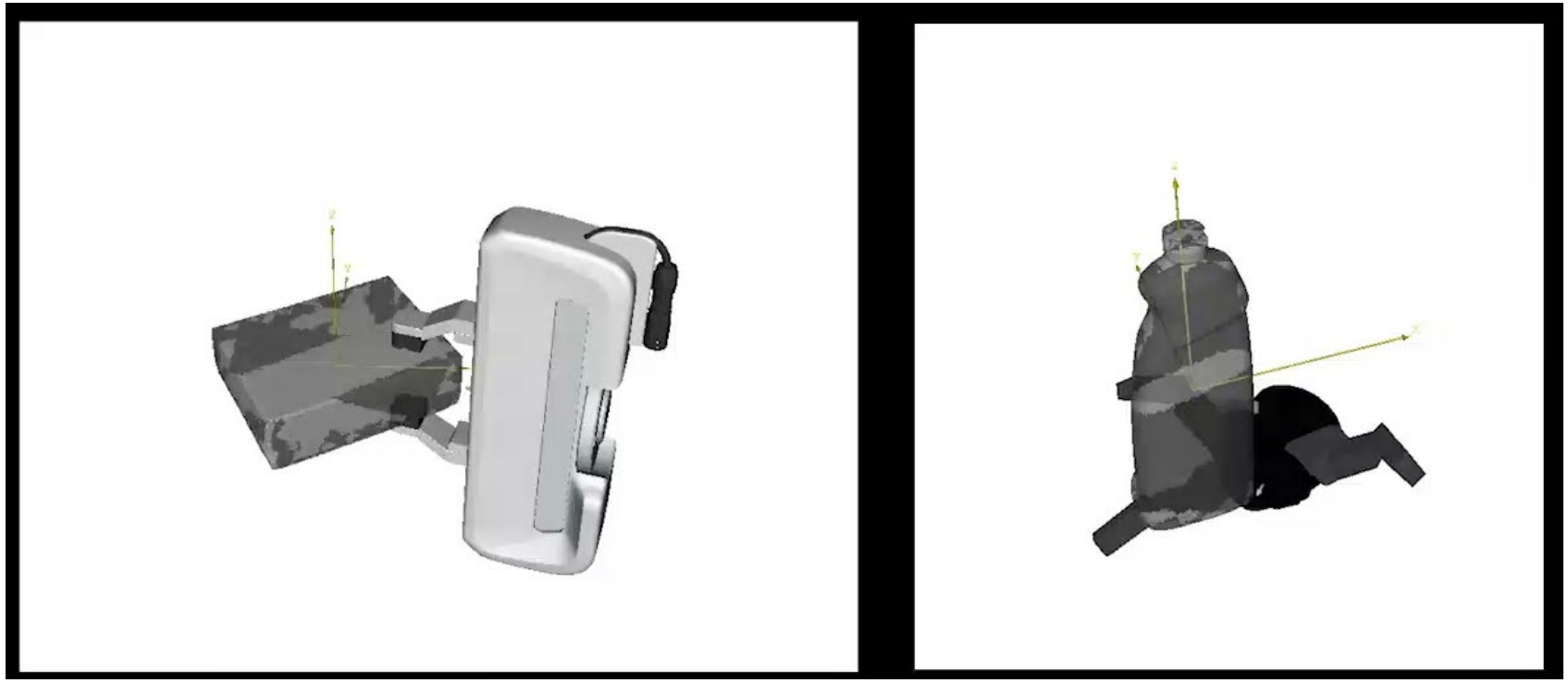
Grasp Planning

CS 6341 Robotics

Instructor Yu Xiang

The University of Texas at Dallas

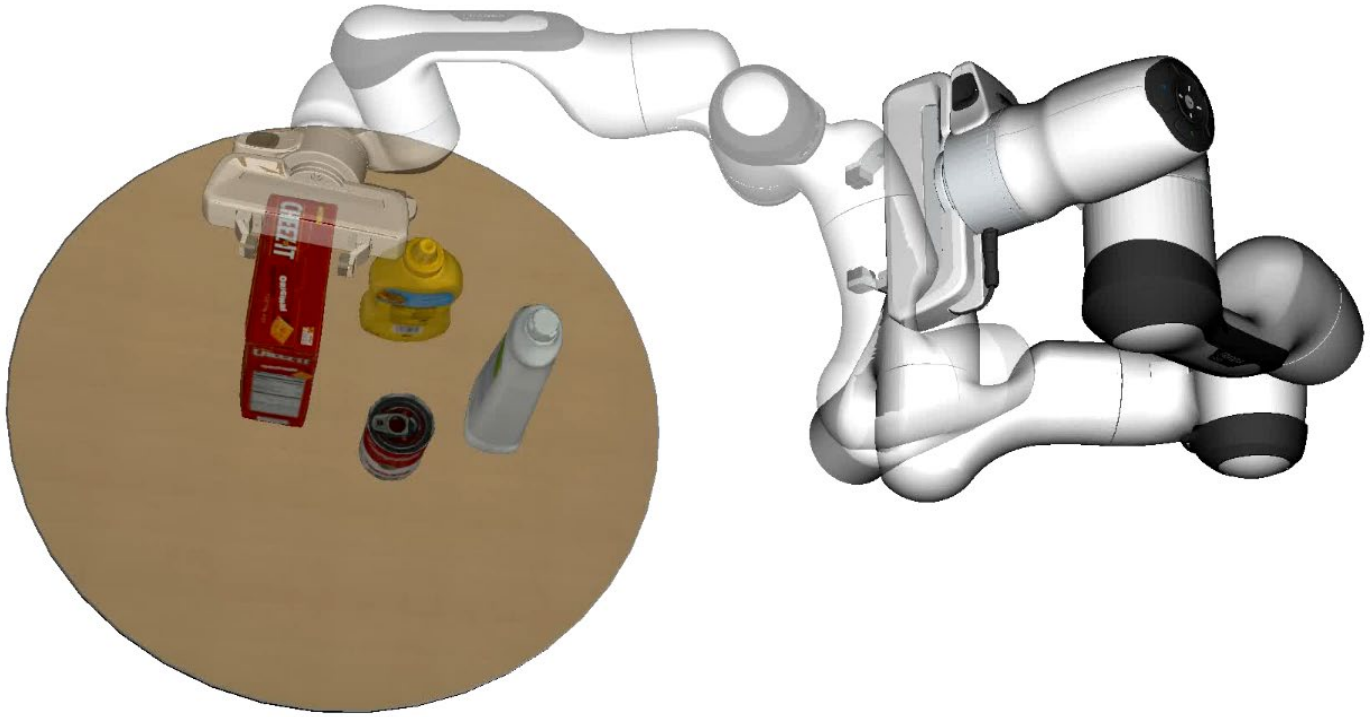
Introduction



GraspIt! https://graspit-simulator.github.io/

Andrew Miller and Peter K. Allen. "Graspit!: A Versatile Simulator for Robotic Grasping". IEEE Robotics and Automation Magazine, V. 11, No.4, Dec. 2004, pp. 110-122.

Grasps are used as the goals for motion planning



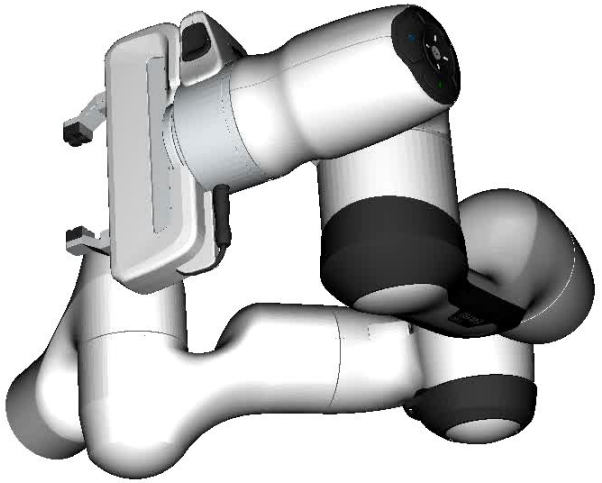
Covariant Hamiltonian Optimization for Motion Planning (CHOMP): Ratliff-Zucker-Bagnell-Srinivasa, ICRA'09

Grasps are used as the goals for motion planning

OMG Iter: 50

100 grasps





Modeling the goal set distribution

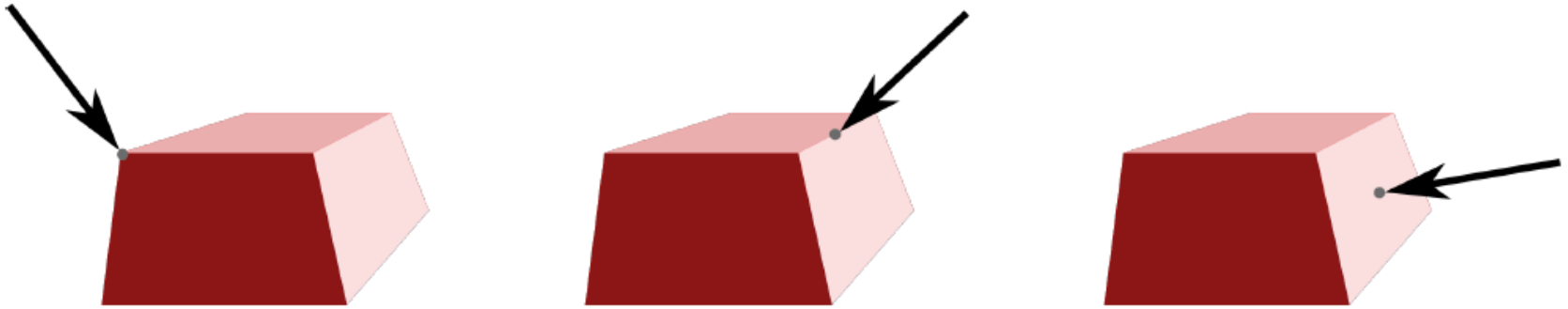
Wang-Xiang-Fox, Manipulation Trajectory Optimization with Online Grasp Synthesis and Selection RSS'20

Grasp Planning

- Model-based methods
  - Force closure analysis
  - Physics simulation

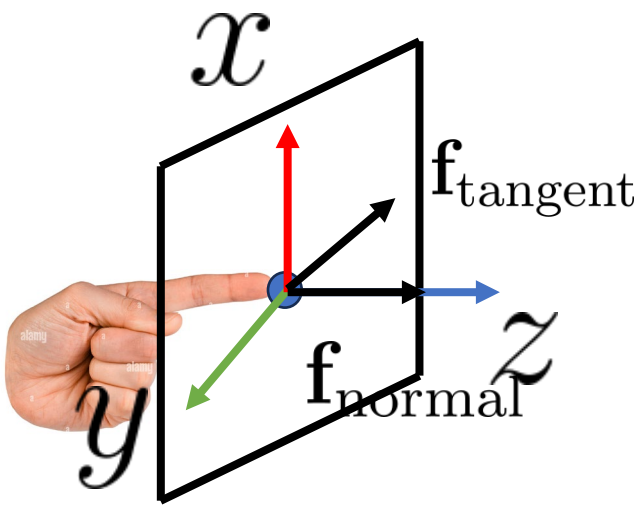
- Learning-based methods
  - Learning generative models with grasping datasets
  - Predicting contact maps, keypoints on object surfaces
  - Predicting hand configurations

Contact Modeling



Point-on-point Point-on-line Point-on-plane

Point-on-Plane Contact Models



$$\mathbf{f} = \mathbf{f}_{\text{normal}} + \mathbf{f}_{\text{tangent}}$$

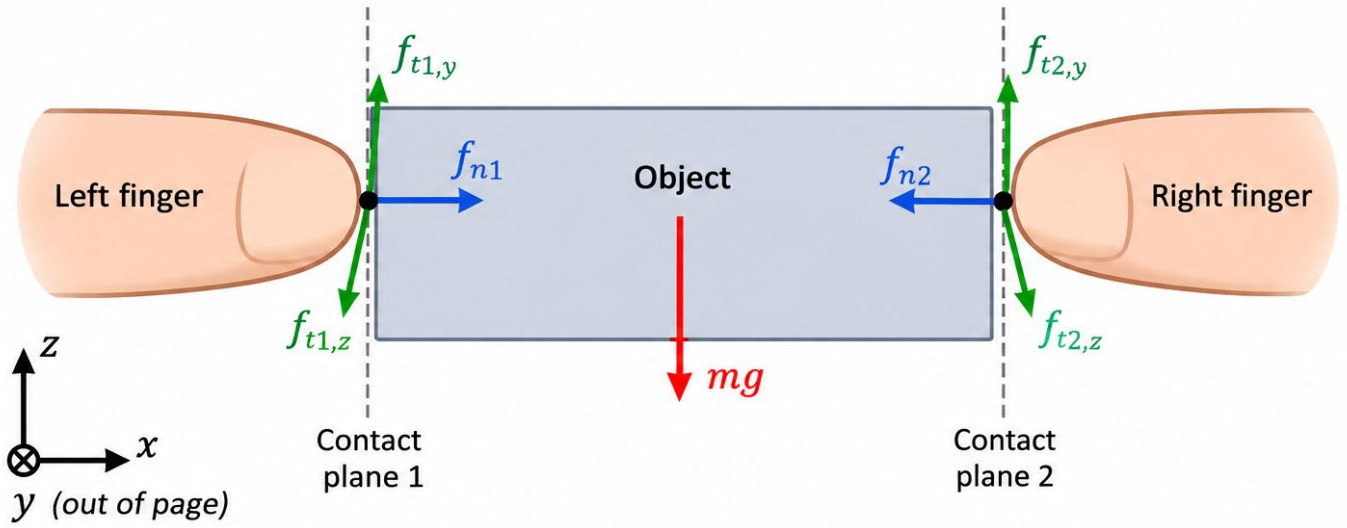
$$\mathbf{f}_{\text{normal}} = [0, 0, f_z]^T \quad f_z \geq 0$$

Acts perpendicular to the contact surface. It is the force that presses the finger against the object, caused by contact pressure

$$\mathbf{f}_{\text{tangent}} = [f_x, f_y, 0]^T$$

Acts parallel to the contact surface. It is associated with friction and allows the finger to resist sliding, caused by friction

Point-on-Plane Contact Models



Left contact forces

f_{n1} : normal force (pushes right)

$f_{t1,y}$: tangential friction (vertical)

$f_{t1,z}$: tangential friction (out of page)

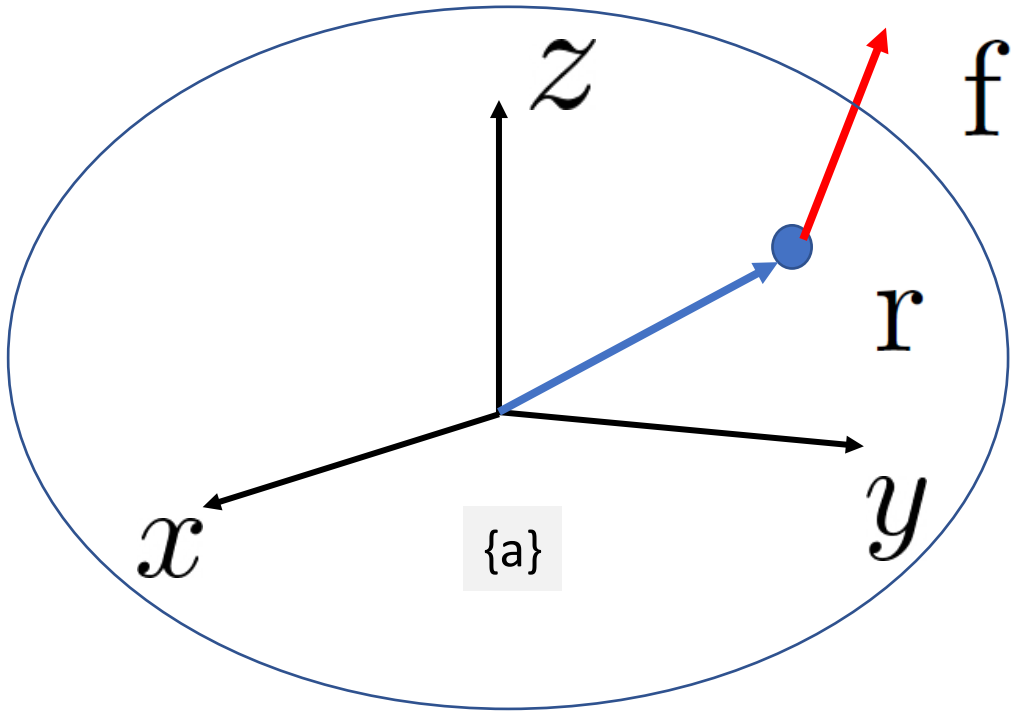
Right contact forces

f_{n2} : normal force (pushes left)

$f_{t2,y}$: tangential friction (vertical)

$f_{t2,z}$: tangential friction (out of page)

Recall Torque and Wrench



A rigid body

Point $r_a \in \mathbb{R}^3$

Force $f_a \in \mathbb{R}^3$

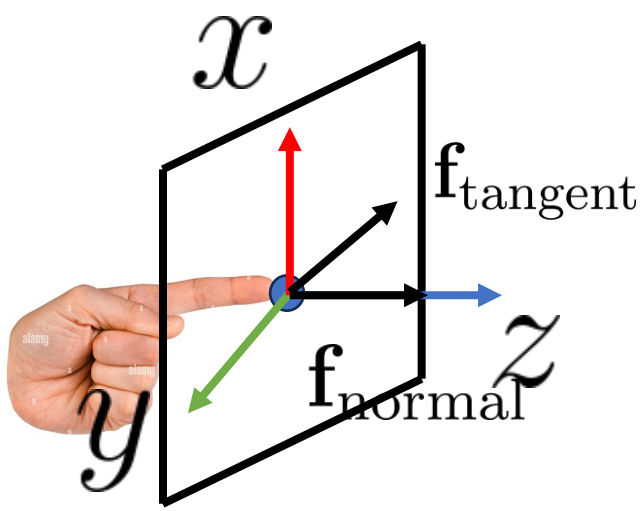
Torque or Moment

$$m_a \in \mathbb{R}^3$$

$$m_a = r_a \times f_a$$

Wrench $$\mathcal{F}_a = \begin{bmatrix} m_a \\ f_a \end{bmatrix} \in \mathbb{R}^6$$

Three Point-on-Plane Contact Models



1. Frictionless Point Contact:

$$\mathcal{F} = \{\mathbf{f}_{\text{normal}} | f_z \geq 0\}$$

$f_x = f_y = 0$

Admissible forces (i.e. forces that don't lead to slipping)

You cannot exert a tangential force on the wall through friction

2. Friction Point Contact:

$$\mathcal{F} = \{\mathbf{f} | \|\mathbf{f}_{\text{tangent}}\| \leq \mu_s \|\mathbf{f}_{\text{normal}}\|, f_z \geq 0\}$$

μ_s static friction coefficient, describes how strongly two contacting surfaces can resist sliding relative to each other.

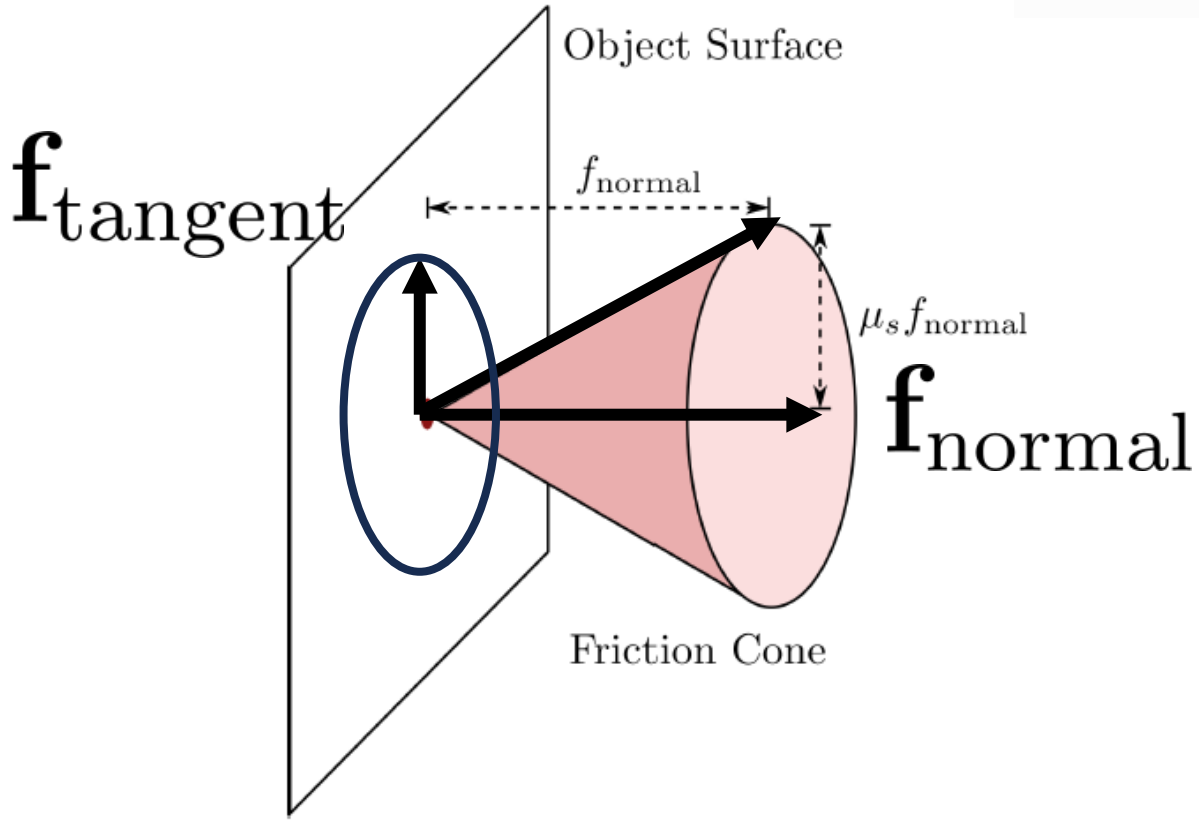
The tangential force is limited because friction has finite capacity

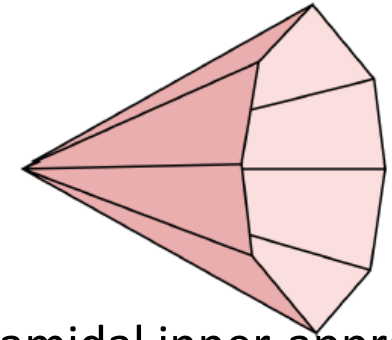
3. Soft-finger Contact Model

$$\mathcal{F} = \{(\mathbf{f}, \tau_{\text{normal}}) | \|\mathbf{f}_{\text{tangent}}\| \leq \mu_s \|\mathbf{f}_{\text{normal}}\|, f_z \geq 0, |\tau_{\text{normal}}| \leq \gamma f_z\}$$

Friction Cone

Contact force	Interpretation
Inside the cone	Static friction can support the force without sliding
On the cone boundary	Friction is at its maximum; slipping may be imminent
Outside the cone	Static friction cannot support the required force; sticking is not possible





A pyramidal inner-approximation of the friction cone

$$\{\boldsymbol{f}_{i,1}, \boldsymbol{f}_{i,2}, \cdots \boldsymbol{f}_{i,m}\}$$

$$\boldsymbol{f}_i = \sum_{j=1}^{m} \alpha_{i,j} \boldsymbol{f}_{i,j}, \quad \alpha_{i,j} \geq 0$$

$$\sum_{j=1}^{m} \alpha_{i,j} \leq 1$$

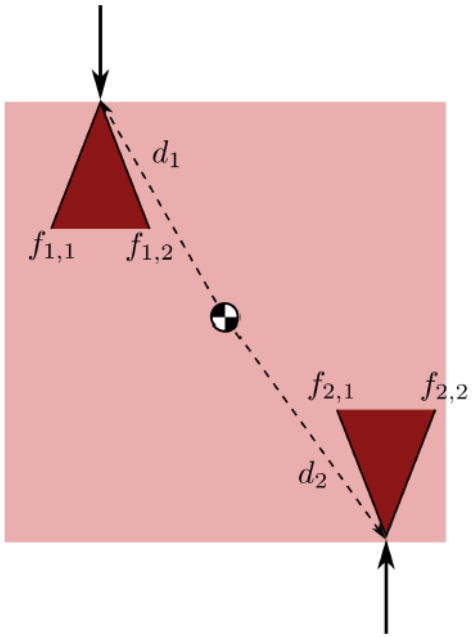
$$\mathcal{F} = \{\mathbf{f} | \|\mathbf{f}_{\text{tangent}}\| \leq \mu_s \|\mathbf{f}_{\text{normal}}\|, f_z \geq 0\}$$

$$\tan(\theta) = \mu_s$$

Grasp Wrench Space

- Wrench: force and torque $w = \begin{bmatrix} f \\ \tau \end{bmatrix} \in \mathbb{R}^6$

- Each contact point applies a wrench to the object

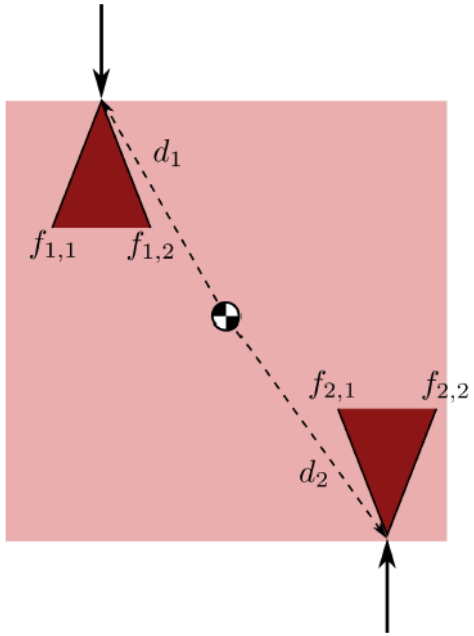


$$\boldsymbol{w}_i = \begin{bmatrix} \boldsymbol{f}_i \\ \lambda(\boldsymbol{d}_i \times \boldsymbol{f}_i) \end{bmatrix}$$

lambda scales the torque components relative to the force components.

$$\omega_i = \begin{bmatrix} I \\ \lambda[\mathbf{d}_i] \end{bmatrix} \mathbf{f}_i \qquad \omega_i = G_i \mathbf{f}_i$$

Grasp Wrench Space



Total wrench on the object

k contact points

$$\boldsymbol{w} = \sum_{i=1}^{k} G_i \boldsymbol{f}_i = G \begin{bmatrix} \boldsymbol{f}_1 \\ \vdots \\ \boldsymbol{f}_k \end{bmatrix}, \quad G = \begin{bmatrix} G_1 & \dots & G_k \end{bmatrix}$$

grasp map

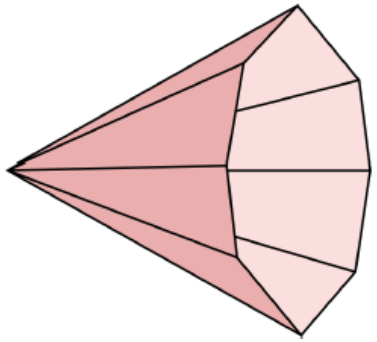
Grasp Wrench Space

- The grasp wrench space W for a grasp with k contact points is the set of all possible wrenches **w** that can be applied to the object through admissible forces:

$$\mathcal{W} := \{\boldsymbol{w} \mid \boldsymbol{w} = \sum_{i=1}^{k} G_i \boldsymbol{f}_i, \quad \boldsymbol{f}_i \in \mathcal{F}_i, \quad i = 1, \ldots, k\}.$$

- If the grasp wrench space is large, the grasp can compensate for a bigger set of external wrenches that might be applied to the object, leading to a more robust grasp

Grasp Wrench Space



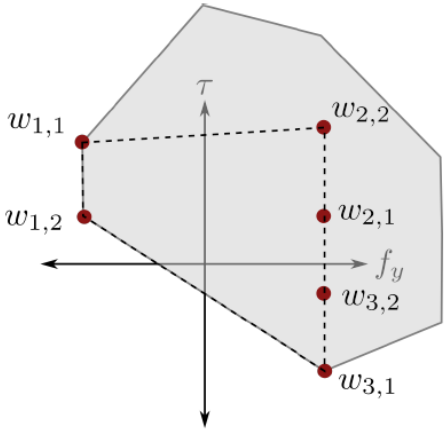
$$\{\boldsymbol{f}_{i,1}, \boldsymbol{f}_{i,2}, \cdots \boldsymbol{f}_{i,m}\}$$

$$\boldsymbol{f}_i = \sum_{j=1}^{m} \alpha_{i,j} \boldsymbol{f}_{i,j}, \quad \alpha_{i,j} \geq 0$$

$$\sum_{j=1}^{m} \alpha_{i,j} \leq 1$$

$$\mathcal{W} = \{\boldsymbol{w} \mid \boldsymbol{w} = \sum_{i=1}^{k} \boldsymbol{w}_i, \quad \boldsymbol{w}_i = \sum_{j=1}^{m} \alpha_{i,j} \boldsymbol{w}_{i,j}, \quad \boldsymbol{w}_{i,j} = \begin{bmatrix} \boldsymbol{f}_{i,j} \\ \lambda(\boldsymbol{d}_i \times \boldsymbol{f}_{i,j}) \end{bmatrix},$$

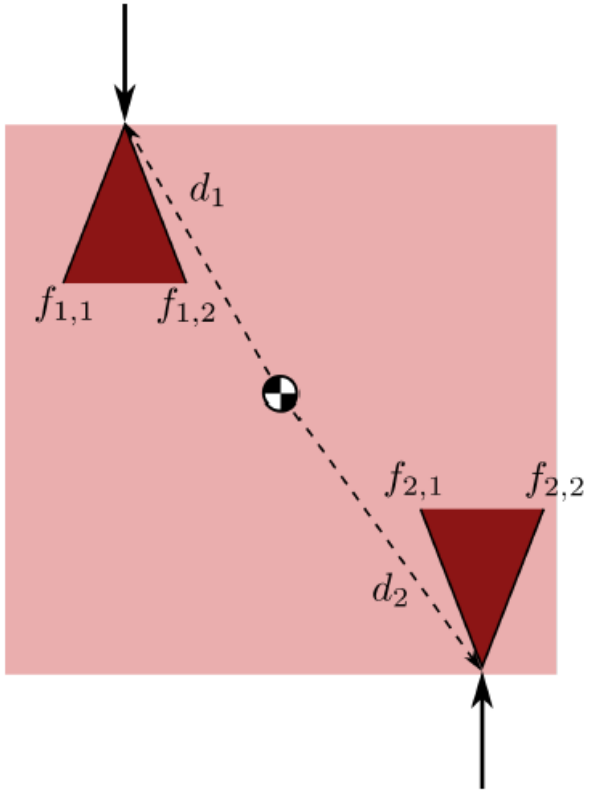
$$\sum_{j=1}^{m} \alpha_{i,j} \leq 1, \quad \alpha_{i,j} \geq 0\}.$$



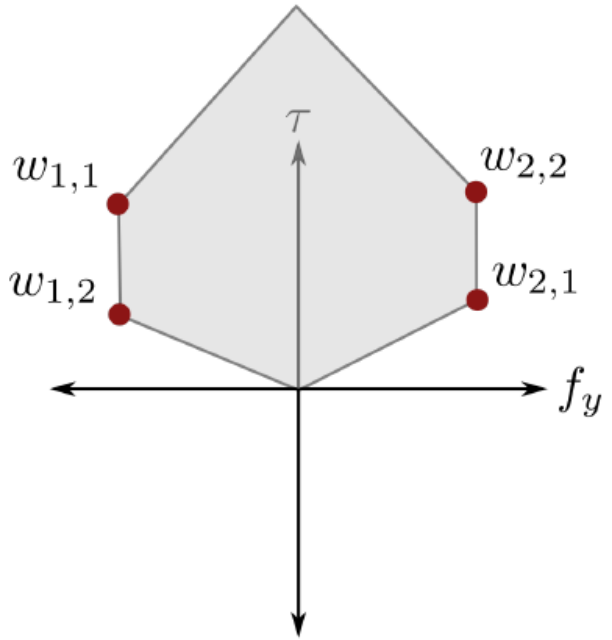
Grasp Wrench Hull

$$\tilde{\mathcal{W}} = \{\boldsymbol{w} \mid \boldsymbol{w} = \sum_{i=1}^{k} \sum_{j=1}^{m} \alpha_{i,j} \boldsymbol{w}_{i,j}, \quad \boldsymbol{w}_{i,j} = \begin{bmatrix} \boldsymbol{f}_{i,j} \\ \lambda(\boldsymbol{d}_i \times \boldsymbol{f}_{i,j}) \end{bmatrix}, \quad \sum_{i=1}^{k} \sum_{j=1}^{m} \alpha_{i,j} = 1, \quad \alpha_{i,j} \geq 0\}$$

Grasp Wrench Space



- f_y: Net force in the y-direction.
- τ: Torque about the object's center.



The grasp wrench space is constructed by combining the wrenches available at all contacts. Its shape depends on contact locations, friction cones, and force constraints. It is not generally a rectangle.

ignoring the horizontal force components f_x

Force Closure Grasp

- A grasp is a force closure grasp if for any external wrench **w** there exist contact forces $\{f_i\}_{i=1}^k$ such that

$$-w = \sum_{i=1}^{k} G_i f_i. \quad f_i \in \mathcal{F}_i, \quad i = 1, \ldots, k,$$

Or

$$-w \in \mathcal{W}.$$

resist any external wrench

The grasp wrench space needs to be $\mathcal{W} = \mathbb{R}^n$

For 2D objects n = 3 and for 3D n = 6

Force Closure Grasp

Grasp Wrench Hull

$$\tilde{\mathcal{W}} = \{\boldsymbol{w} \mid \boldsymbol{w} = \sum_{i=1}^{k}\sum_{j=1}^{m} \alpha_{i,j}\boldsymbol{w}_{i,j}, \quad \boldsymbol{w}_{i,j} = \begin{bmatrix} \boldsymbol{f}_{i,j} \\ \lambda(\boldsymbol{d}_i \times \boldsymbol{f}_{i,j}) \end{bmatrix}, \quad \sum_{i=1}^{k}\sum_{j=1}^{m} \alpha_{i,j} = 1, \quad \alpha_{i,j} \geq 0\}$$

- Theorem: In an n-dimensional vector space with

$$\mathcal{W} := \{\boldsymbol{w} \mid \boldsymbol{w} = \sum_{k=1}^{N} \beta_k \boldsymbol{w}_k, \quad \beta_k \geq 0\},$$

$\mathcal{W} = \mathbb{R}^n$ if and only if the set $\{\boldsymbol{w}_k\}_{k=1}^{N}$ contains at least $n+1$ vectors, n of the vectors are linearly independent, and there exists scalars $\beta_k > 0$ such that:

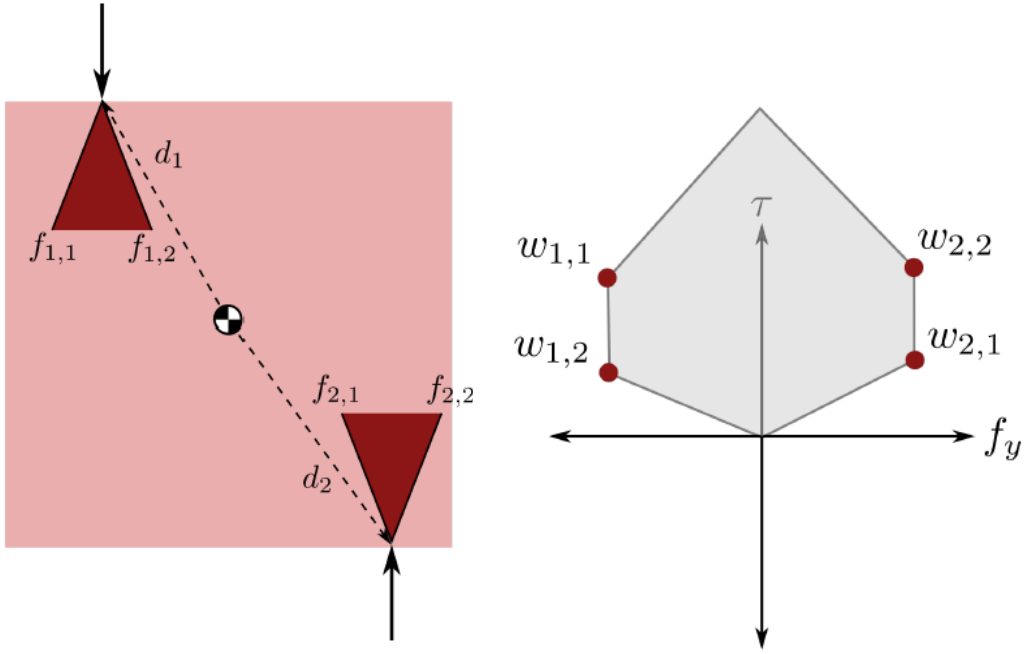
$$\sum_{k=1}^{N} \beta_k \boldsymbol{w}_k = 0.$$

grasp wrench space W must contain the origin in its interior

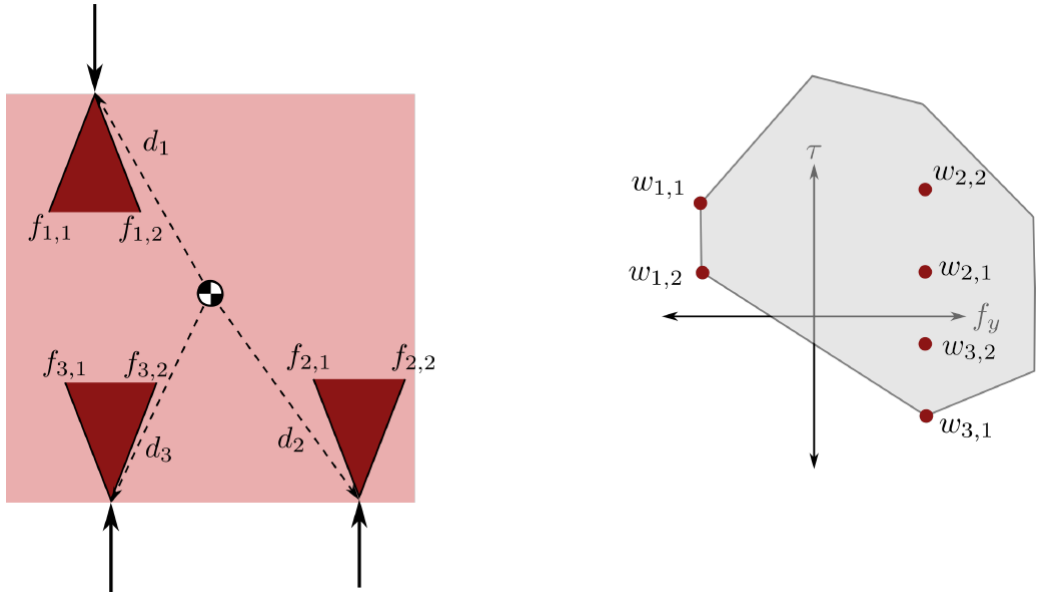
Force closure concerns the directions of external wrenches that a grasp can resist. A bounded grasp wrench space accounts for their magnitudes as well.

Force Closure Grasp

n = 3



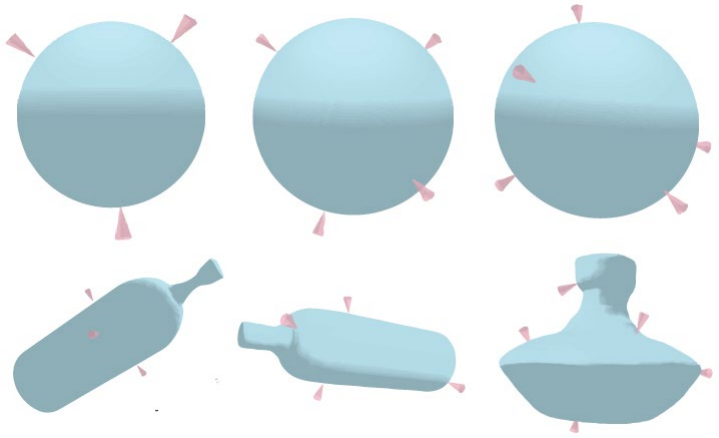
Not force closure



force closure

Optimization to find force closure grasps

- Find contact points $\mathbf{x}_1, \ldots, \mathbf{x}_k$, and contact forces $\mathbf{f}_1, \ldots, \mathbf{f}_k$



$$GG' \succeq \epsilon I_{6\times 6},$$
$$Gf = 0,$$
$$f_i^T c_i > \frac{1}{\sqrt{\mu^2+1}}|f_i|,$$
$$x_i \in S,$$

G is full rank

grasp wrench hull contains the origin in its interior

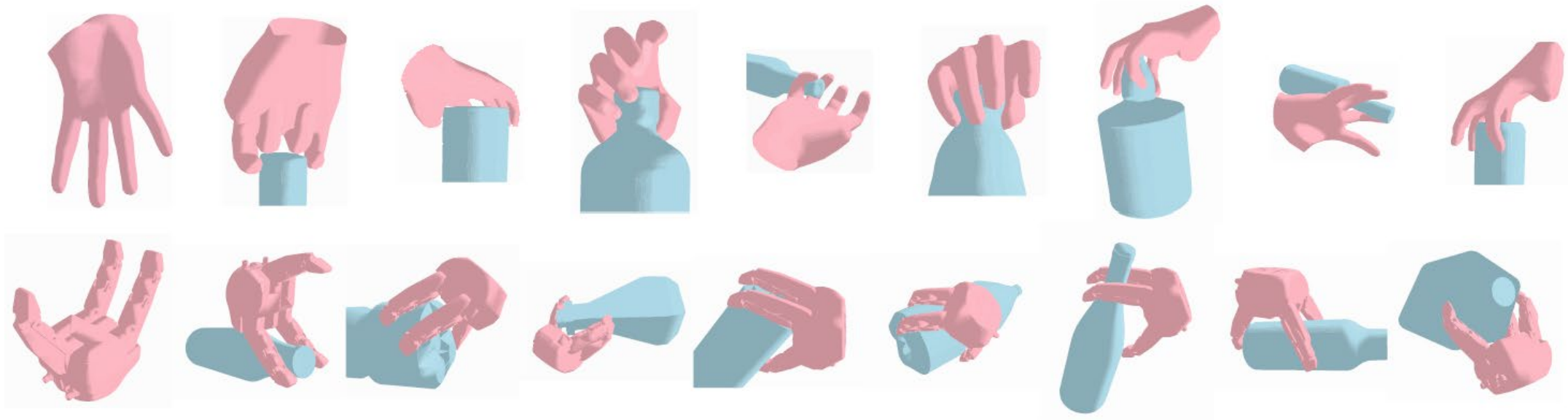
Friction cone constraint

Point on object surface

$$G = \begin{bmatrix} I_{3\times 3} & I_{3\times 3} & \cdots & I_{3\times 3} \\ \lfloor x_1 \rfloor_\times & \lfloor x_2 \rfloor_\times & \cdots & \lfloor x_n \rfloor_\times \end{bmatrix}$$

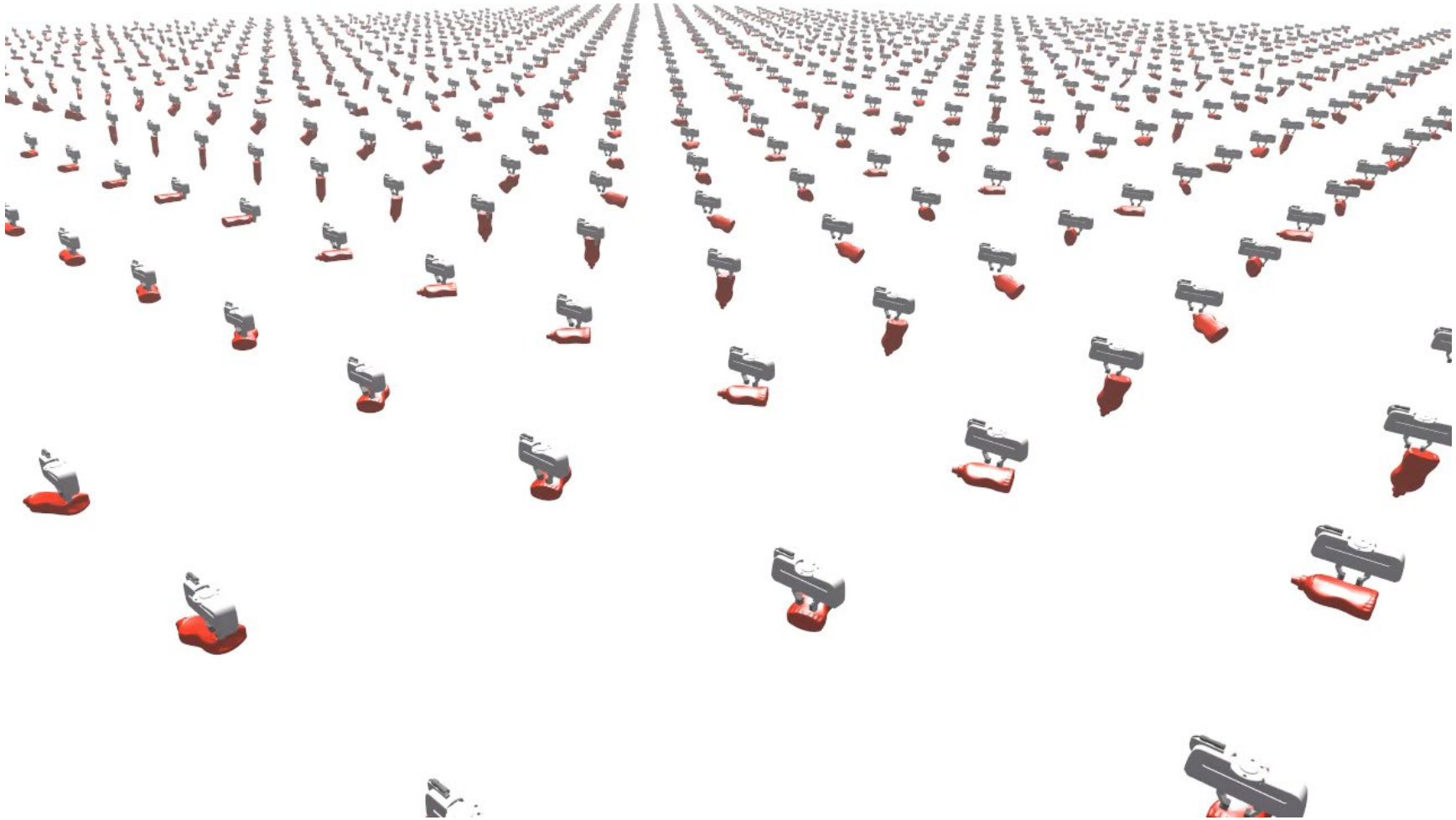
[1] Synthesis and Optimization of Force Closure Grasps via Sequential Semidefinite Programming. Hongkai Dai, Anirudha Majumdar and Russ Tedrake, 2015
[2] Synthesizing Diverse and Physically Stable Grasps With Arbitrary Hand Structures Using Differentiable Force Closure Estimator. Tengyu Liu , Zeyu Liu, Ziyuan Jiao , Yixin Zhu , and Song-Chun Zhu, 2022

Optimization to find force closure grasps



Synthesizing Diverse and Physically Stable Grasps With Arbitrary Hand Structures Using Differentiable Force Closure Estimator. Tengyu Liu , Zeyu Liu, Ziyuan Jiao , Yixin Zhu , and Song-Chun Zhu, 2022

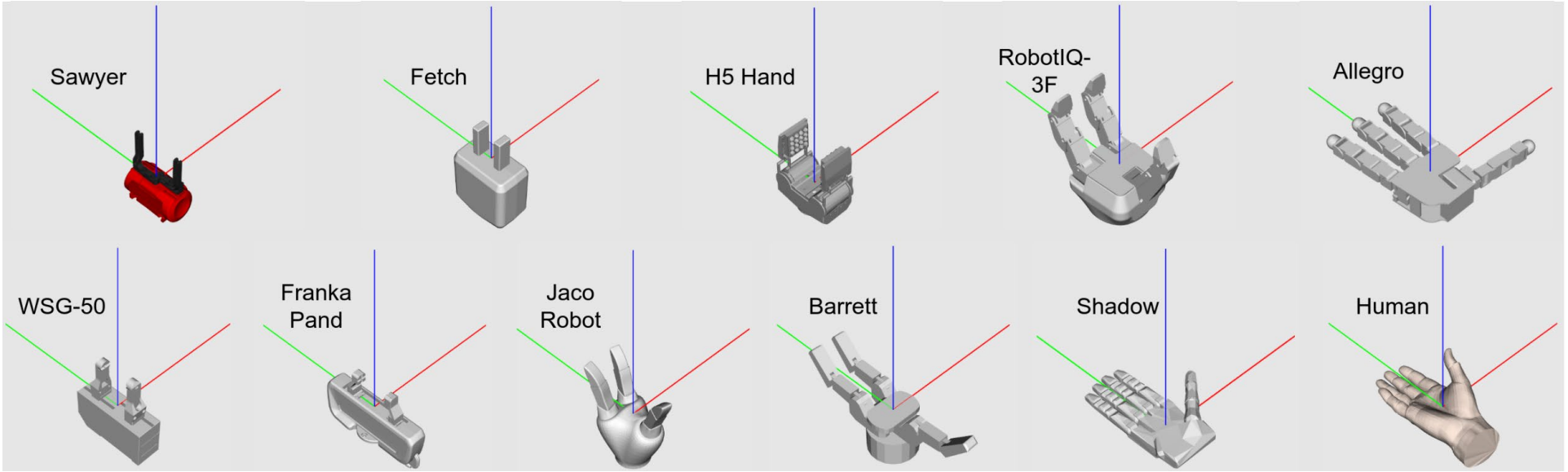
Use Physics Simulation for Grasp Planning





Eppner-Mousavian-Fox, ISRR'19

MultiGripperGrasp

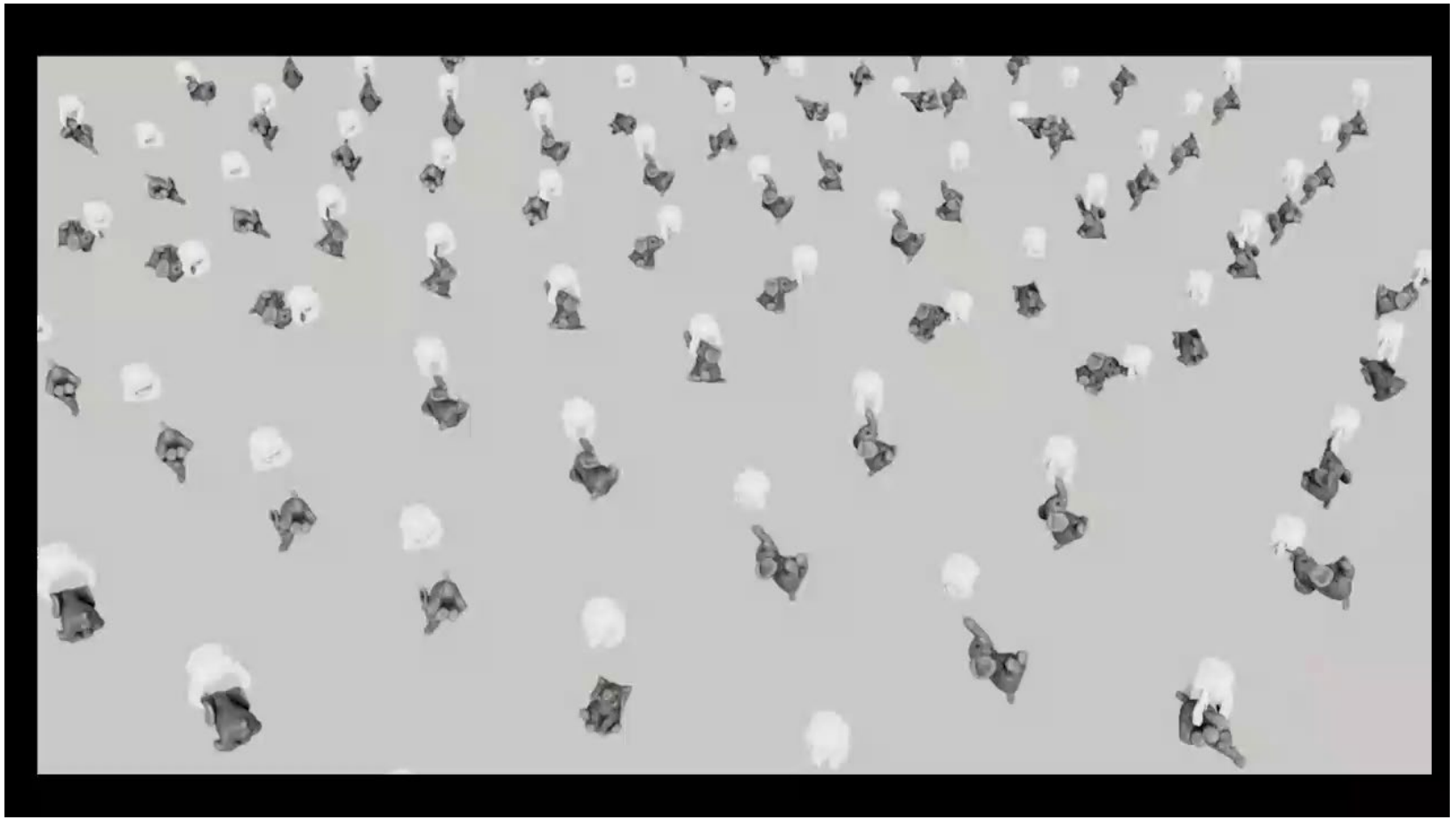


- 11 grippers (aligned with palm directions)
  - 2-finger grippers: Fetch, Franka Panda, WSG50, Sawyer, H5 Hand
  - 3-finger grippers: Barrett, Robotiq-3F, Jaco Robot
  - 4-finger grippers: Allegro
  - 5 finger grippers: Shadow, Human Hand

MultiGripperGrasp: A Dataset for Robotic Grasping from Parallel Jaw Grippers to Dexterous Hands. Luis Felipe Casas Murrilo*, Ninad Khargonkar*, Balakrishnan Prabhakaran, Yu Xiang (*equal contribution). In IROS, 2024.

MultiGripperGrasp

- Generate initial grasps using GraspIt! Ranking grasps in Isaac Sim



MultiGripperGrasp: A Dataset for Robotic Grasping from Parallel Jaw Grippers to Dexterous Hands. Luis Felipe Casas Murrilo*, Ninad Khargonkar*, Balakrishnan Prabhakaran, Yu Xiang (*equal contribution). In IROS, 2024.

Model-based Grasping vs Model-free Grasping

SceneReplica, ICRA'24: https://irvlutd.github.io/SceneReplica/



Input real world scene



6D Pose Estimation



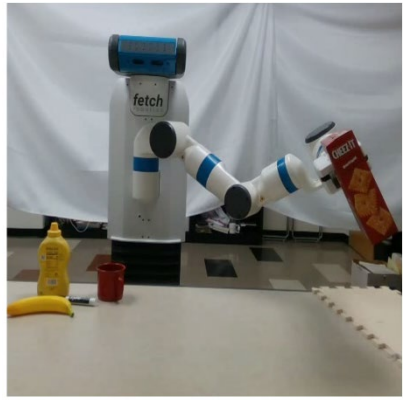
Offline Grasp Database



Motion Planning Setup



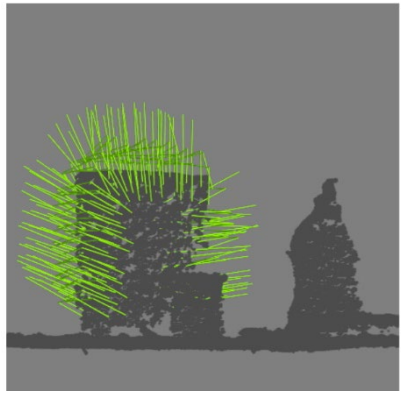
Grasping & Lifting



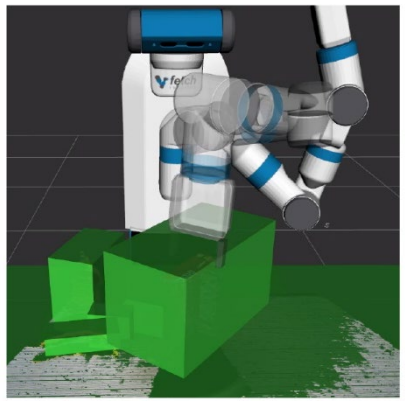
Moving arm for Dropoff



Unseen Object Segmentation



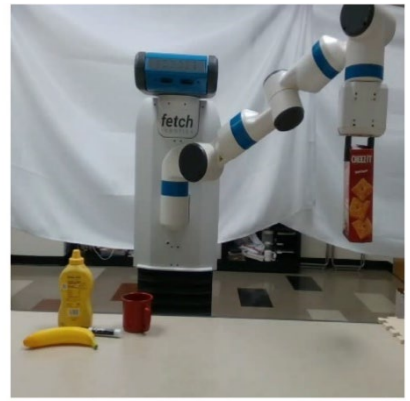
Model-free Grasp Planning



Motion Planning Setup

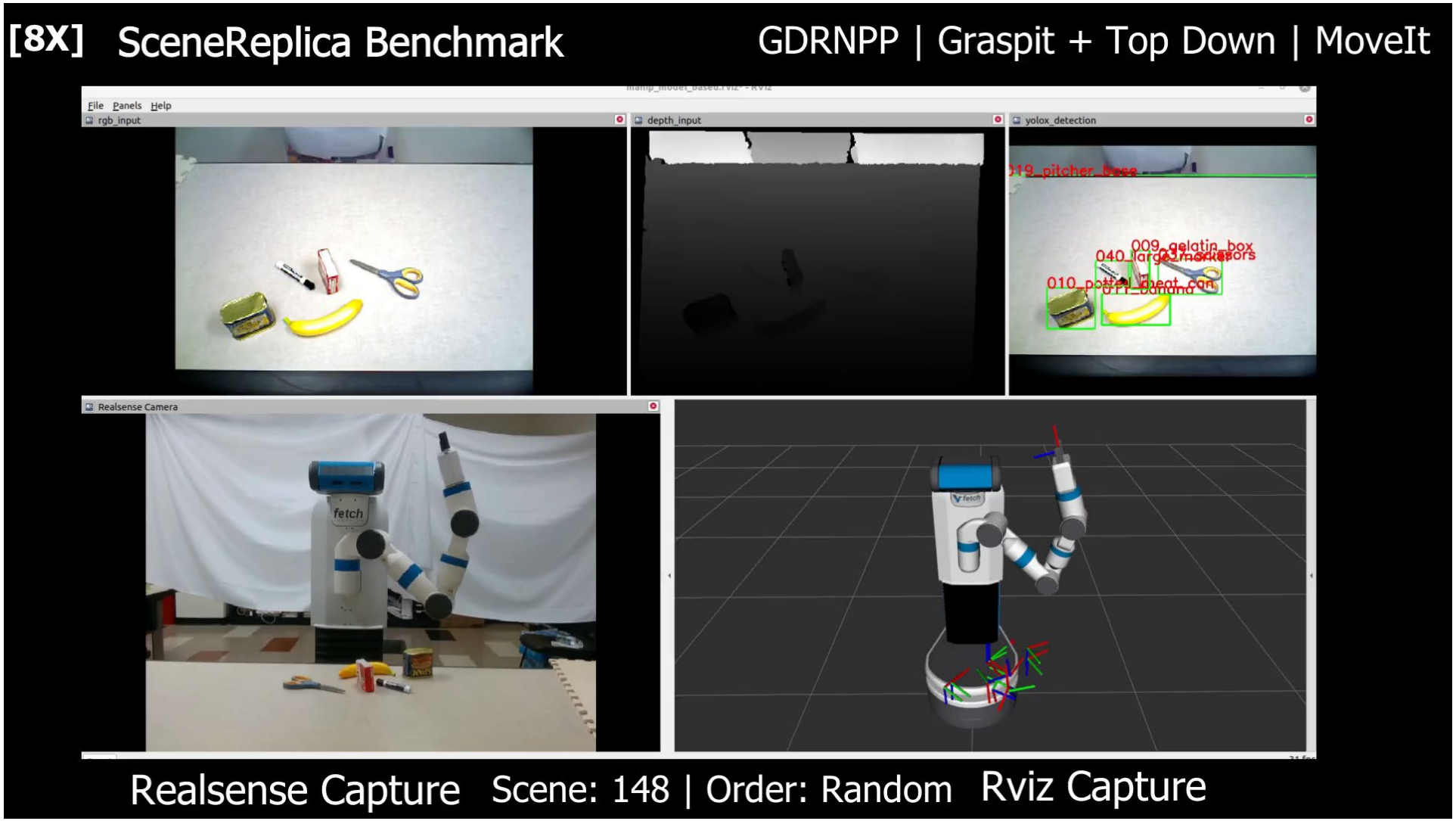


Grasping & Lifting



Moving arm for Dropoff

Model-based Grasping Example



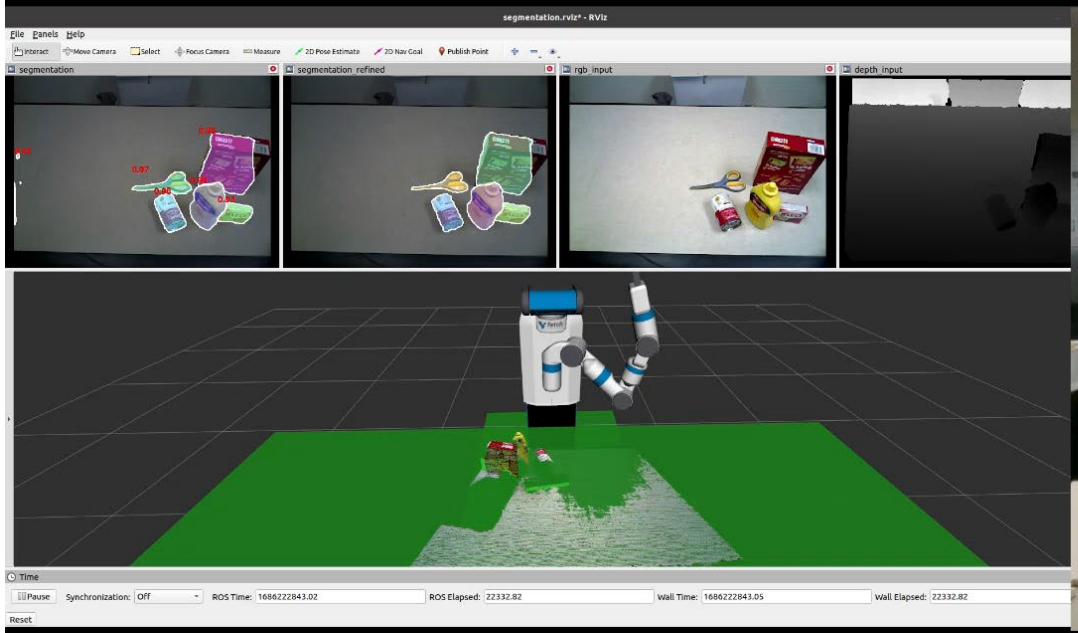
Model-free Grasping Example

8X

SceneReplica Benchmark

MSMFormer | Contact GraspNet + Top Down | MoveIt

Scene: 130 | Order: Random





Rviz Capture

Realsense Capture

Summary

- Point-on-Plane Contact Models
- Friction cone
- Grasp Wrench Space
- Force Closure Grasp
- Optimization to find force closure grasps

Further Reading

[1] Main reference: Stanford Principles of Robot Autonomy-II: Grasping lectures

[2] Friction Cone

[3] Classical Grasp Planning: GraspIt Simulator

[4] Dex-Net 2.0: Deep Learning to Plan Robust Grasps with Synthetic Point Clouds and Analytic Grasp Metrics

[5] 6-DOF GraspNet: Variational Grasp Generation for Object Manipulation

[6] SceneReplica: https://irvlutd.github.io/SceneReplica/